Academic Typesetting Portfolio

Thesis-layout demonstration

Temperature and Power in a Synthetic Sensor Dataset

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LaTeX document design and typesetting sample

Portfolio demonstration only. The dataset is synthetic; this document is not an academic submission and is not affiliated with a university.

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Abstract

This thesis-layout sample uses 2,880 hourly observations from three example monitoring sites. The data are synthetic and generated by a fixed-seed Python script. The document shows how to organize a long LaTeX project, state a regression model, include diagrams and figures, and manage references and appendices. The fitted hinge model has an in-sample coefficient of determination of 0.850 on the generated data. This value describes the fit to these rows; it is not a test of performance on new data. The control-system figures are separate illustrations and are not tests on the dataset.

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Chapter 1

Introduction

1.1 Motivation

Energy-monitoring systems record measurements over time. Temperature can be one of several factors associated with power demand. This sample uses synthetic data and generated hourly data to show how such measurements can be summarized and typeset. The values are illustrative; they do not come from a live site or a published experiment.

1.2 Questions

The example is organized around three questions:

RQ1. How are temperature and power associated in the generated data?

RQ2. How do the three example sites differ in their generated power values?

RQ3. How can a feedback-control system be illustrated with a block diagram and a step-response plot?

1.3 Structure

Chapter 2 describes the software and background, Chapter 3 explains the data and model, and Chapter 4 reports the summaries produced by the scripts. Chapter 5 states what these examples do and do not show. The commands used to regenerate the files are listed in Chapter A.

Chapter 2

Background and Tools

2.1 Scientific Python

The data-generation and plotting scripts use pandas for tabular data [4], Matplotlib [2], and seaborn [7]. The charts use a colour palette designed to remain distinguishable for readers with common forms of colour-vision deficiency [8].

2.2 Attention Models

The Transformer architecture was introduced by Vaswani et al. [6]. Figure 2.1 is a simplified encoder diagram drawn with TikZ. It is included as a vector-graphics example, not as part of the power-demand model.

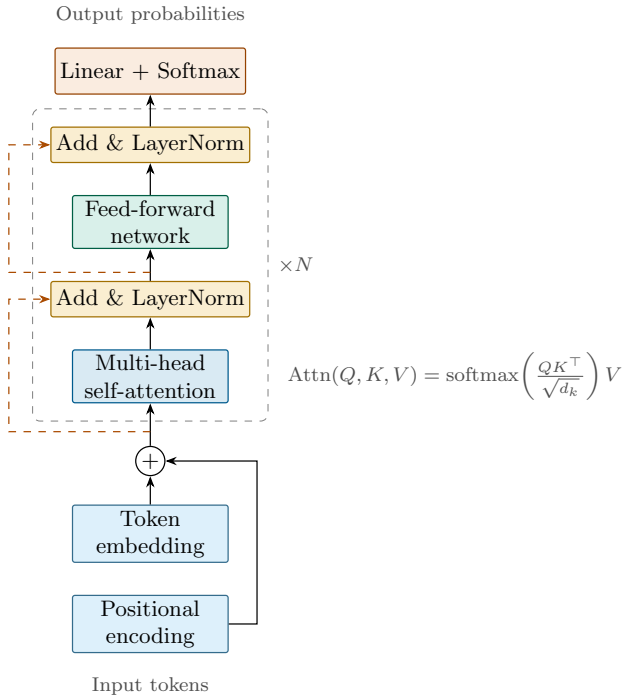


Figure 2.1: Simplified Transformer encoder stack with residual connections.

2.3 Least Squares

For a full-rank design matrix, ordinary least squares has a closed-form estimator. Its expectation and covariance follow from the linear model assumptions; a standard treatment is given by Kay [3]. The proposition in Proposition 3.1 states these estimator properties. It does not select a particular regression model.

Definition 2.1 (Hinge feature)**.** *For a threshold τ, the hinge feature of temperature T is* $h(T) = \max(T - \tau, 0)$.

2.4 Feedback Control

A proportional–integral–derivative (PID) controller is a standard feedback-control structure; Ogata provides an introduction [5]. A finite-state machine can represent supervisory mode changes [1]. The diagram in Figure 2.2 is a small illustrative state machine.

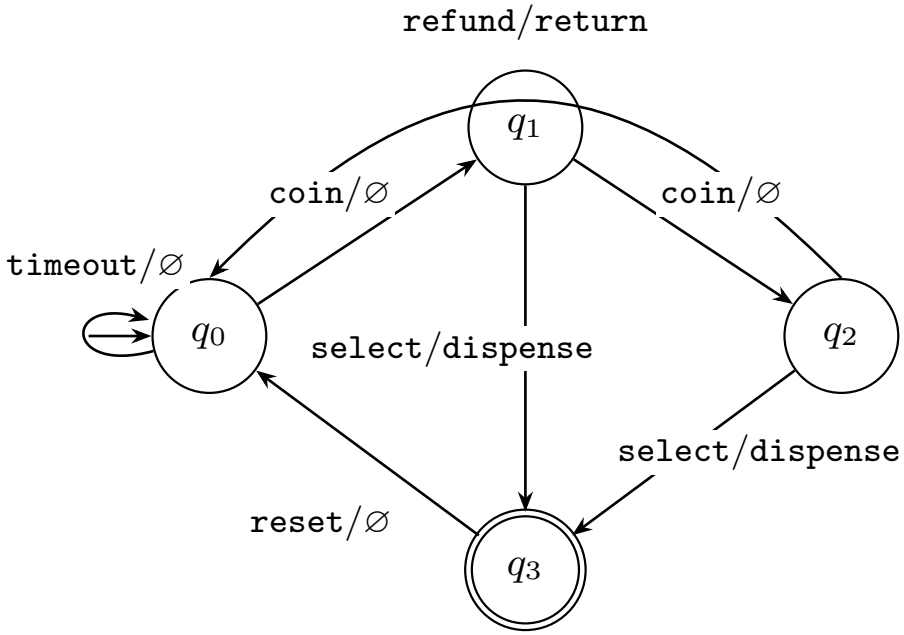


Figure 2.2: Example finite-state machine with labelled transitions.

Chapter 3

Methodology

3.1 Generated Data

The script `scripts/make_data.py` creates 2,880 hourly rows for three example sites. The temperature and power columns contain 50 deliberately inserted missing values in total. The figure script fills those gaps by time-ordered linear interpolation. The dataset is synthetic; it is generated to exercise the plotting and typesetting workflow.

3.2 Hinge Regression

The example fits the hinge regression model in Equation (3.1):

$$P_k = \beta_0 + \beta_1 \max(T_k - \tau, 0) + \varepsilon_k, \tag{3.1}$$

where T_k is temperature, P_k is power, and τ is the threshold. The script uses a grid search over candidate thresholds. For each candidate, it solves for β_0 and β_1 by ordinary least squares and keeps the candidate with the lowest residual sum of squares.

Proposition 3.1. *For a full-column-rank design matrix $\boldsymbol{X}$, the ordinary least-squares estimator*

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} \tag{3.2}$$

is unbiased under the zero-mean error assumption. If the errors have covariance $\sigma^2 \boldsymbol{I}$, then $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}$.

3.3 Control-Diagram Examples

Figure 3.1 shows the structure of a feedback loop. The PID equation below defines the controller signal in terms of the error:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \dot{e}(t). \tag{3.3}$$

The second-order step responses in Figure 3.2 compare damping ratios of 0.2, 0.5, and 0.8. The lower-damped response has more overshoot. These plots illustrate control concepts; no controller is fitted to or tested on the generated sensor data.

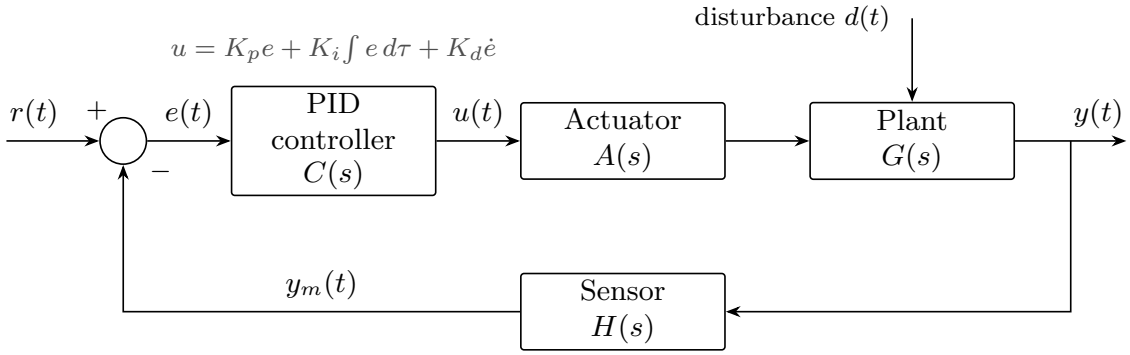


Figure 3.1: Block diagram of a feedback loop with a PID controller.

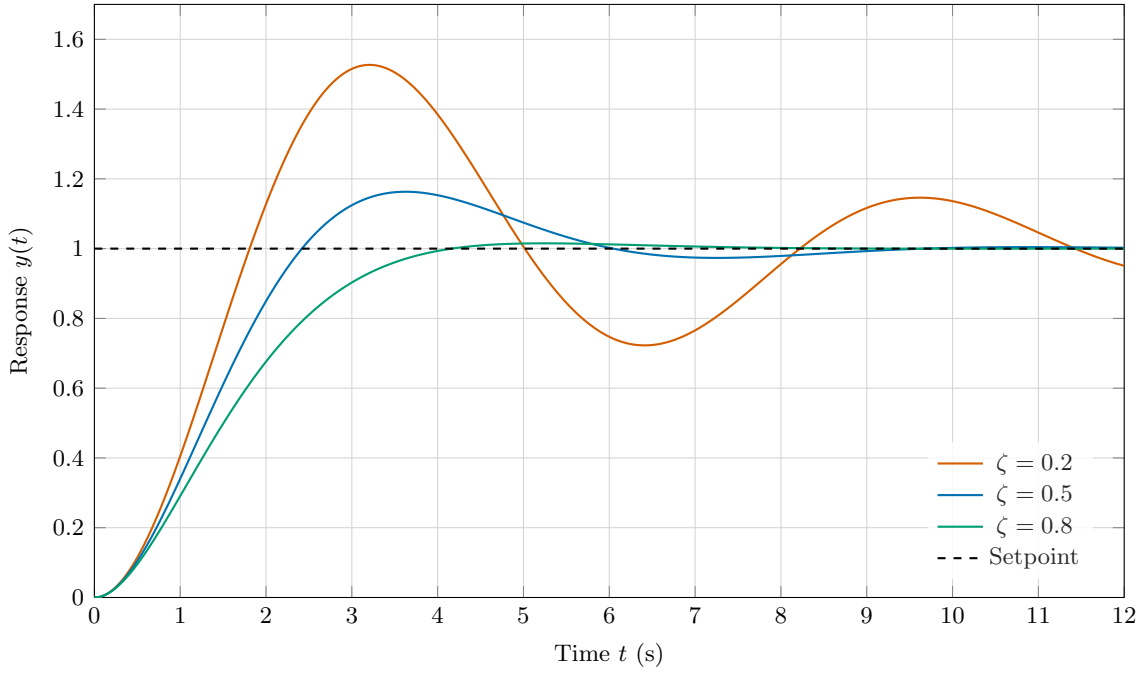


Figure 3.2: Second-order unit-step responses for three damping ratios.

3.4 Reproducible Figures

The charts and summary table are generated by `scripts/generate_figures.py`. The full sequence of commands is given in Chapter A.

Chapter 4

Results

4.1 Exploratory Summaries

Temperature and power have a positive Pearson correlation in this generated dataset ($r = 0.86$; Figure 4.1). The hinge regression has an in-sample coefficient of determination of $R^2 = 0.850$. This is a summary of the same rows used to fit the model, not an estimate of performance on unseen measurements.

Figure 4.1 contains four views of the synthetic data: the daily power series in panel (a), the variable-correlation matrix in panel (b), the site distributions in panel (c), and power against temperature in panel (d). The final panel shows both the fitted hinge curve and a dashed linear baseline. Per-site counts and summary values are listed in Table 4.1.

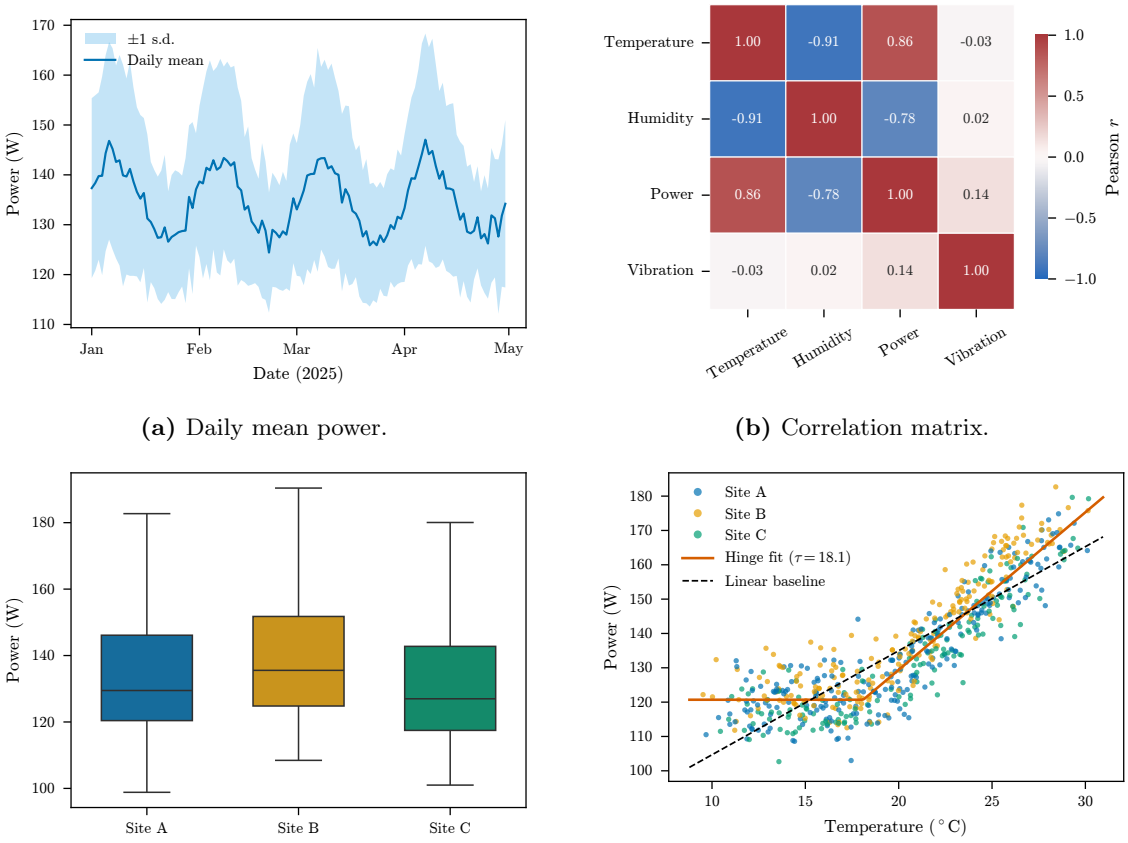
Table 4.1: Summary statistics by example site.

Site	n	$\bar{T}$ (°C)	$\bar{P}$ (W)	s_P (W)	$\bar{a}$ (g)
Site A	1151	19.87	133.74	16.60	0.377
Site B	989	20.18	139.18	16.92	0.385
Site C	740	19.87	131.00	16.21	0.383

4.2 Discussion

Site B has the highest mean power in this generated sample. The data-generation script adds a positive site offset to Site B, so this difference is built into the example. The correlation and R^2 describe the generated rows; they do not establish a causal relationship or validate a forecast for a real site.

Proposition Proposition 3.1 concerns the unbiasedness and covariance of the ordinary least-squares estimator under its stated assumptions. It is a result about the estimator, not evidence that the hinge specification is the correct model for real measurements.



(a) Daily mean power.

(b) Correlation matrix.

(c) Power by site.

(d) Power against temperature.

Figure 4.1: Four summaries generated from the synthetic hourly dataset. Panel (d) shows the fitted hinge curve and the linear baseline.

Chapter 5

Conclusion

The fitted hinge regression has an in-sample R^2 of 0.850 on the generated observations. The data were created from a temperature-based formula with site offsets, so this statistic describes how the fitted curve represents the example data. It is not evidence about real monitoring sites, a causal effect, or accuracy on new observations.

The PID block diagram and the second-order step-response plot illustrate a feedback-control structure and the effect of changing damping. No controller was tuned against the generated dataset, and no physical system was controlled.

The sample's purpose is to show a reproducible LaTeX project: the chapters, diagrams, figures, tables, bibliography, and index are kept in separate files and rebuilt by scripts.

Appendix A

Reproducibility

From the repository root, install the Python requirements and build all three documents. The OLS covariance derivation is in Chapter B:

```
pip install -r requirements.txt
make clean
make all
```

The data are recreated with `python scripts/make_data.py`. The figures and LaTeX tables are recreated with `python scripts/generate_figures.py`. To update the committed PDF samples and JPG previews after a successful build, run `make refresh-samples`.

Appendix B

Derivation of the OLS Covariance

Substituting $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into Equation (3.2) gives

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{\varepsilon}.$$

If $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}$, then

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \tag{B.1}$$

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top (\sigma^2 \boldsymbol{I}) \boldsymbol{X} (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \tag{B.2}$$

$$= \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}. \tag{B.3}$$

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